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Growing independent thinkers: the role of technology-supported cooperative learning in fostering self-directed learning in mathematics

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Equipping learners with the ability to manage their own learning has become a priority in mathematics education, particularly at the lower-secondary level. This resonates with heutagogical calls for self-determined learning from an early age. However, in many under-resourced schools, traditional teacher-centred instruction remains dominant, limiting opportunities for autonomy and collaboration. Cooperative learning (CL) shows promise in supporting engagement and achievement and its potential is strengthened when combined with digital tools. Yet, limited research has explored how such technology-supported cooperative learning (TSCL) interventions, informed by the heutagogical lens, influence Grade 8 learners' perceptions of their self-directed learning (SDL) abilities in mathematics. This study investigated the effect of a TSCL intervention on Grade 8 learners' perceptions of four SDL dimensions: learning motivation, planning and implementation, self-monitoring and interpersonal communication. Conducted in 10 South African public schools, two schools formed the experimental group and eight served as the control group, with 427 learners in Cycle 1 and 522 in Cycle 2. A DBR approach and sequential explanatory mixed-methods design were used. Data were collected through pre/post SDLI questionnaires, interviews and classroom observations. Findings showed statistically significant improvements in all SDL dimensions, especially motivation and interpersonal communication. Qualitative data affirmed increased learner autonomy and engagement. Interpreted through a heutagogical lens, the findings suggest that TSCL can begin to cultivate capabilities associated with self-determined learning such as agency, collaboration and reflection in mathematics classrooms. The study highlights TSCL as an effective strategy for promoting SDL, supported by sustained teacher professional development.

KEYWORDS

21st-century skills, design-based research, mathematics education, self-directed learning, STEM, technology-supported cooperative learning

1 Introduction

In today's fast-changing and often unpredictable world, education systems must prioritise the development of learners for lifelong, self-directed learning (SDL) (Panthalookaran, 2022). In the field of mathematics education, particularly at the lower-secondary level, establishing SDL capabilities becomes essential for learning abstract concepts and for academic achievement (Suhaimee et al., 2025). However, in many schools with limited resources, the traditional, teachercentred classroom is the norm (Petersen et al., 2020), which limits learner

autonomy and engagement. Cooperative learning (CL) has long been recognised for fostering engagement, critical thinking (Johnson et al., 1998) and positive affect in mathematics classrooms (Laubscher and Bosch, 2025). Integrating CL strategies with digital technologies (e.g., virtual manipulatives, LMS, interactive quizzes) can expand on this by guiding learners in managing their own learning and working collaboratively with peers (Morris and Rohs, 2023). Technology-supported cooperative learning (TSCL) is closely aligned with the theoretical framework of Computer Supported Collaborative Learning (CSCL), which investigates how technology mediates cognitive and social interactions to support joint meaning-making (Jeong et al., 2019). The focus of this study is on investigating the use of CL with the five basic principles (Johnson and Johnson, 2009) in a technology-enhanced environment informed by the heutagogical lens.

Despite widespread calls for technology-supported SDL-conducive environments, their implementation in Grade 8 mathematics remains underexplored, particularly regarding how learners perceive their SDL abilities in this context. According to the South African Department of Basic Education's 2022/23 Annual Report, the national learner-teacher ratio remains high. The ratio is currently around 34 learners per teacher, which emphasises the systemic constraints on individualised attention and learner autonomy (Department of Basic Education, 2023). Such overcrowded conditions limit opportunities for self-directed engagement, reinforcing the need for pedagogies that actively develop learner agency. Although numerous studies have explored the role of technology in promoting SDL within different disciplines in Higher Education Institutions (Alanoglu et al., 2025; Laubscher and Bosch, 2025; Mentz and Bailey, 2019), limited research has examined how TSCL specifically influences Grade 8 learners' perceptions of their SDL. Moreover, most research relies solely on self-report or quantitative designs, lacking the triangulation of qualitative classroom observations and interviews essential for validating and deepening quantitative findings.

Heutagogical perspectives further deepen this focus on SDL. Heutagogy, a form of self-determined learning (Blaschke, 2012), positions learners as active agents who make deliberate choices about what and how they learn, building capabilities for lifelong learning in complex, technology-rich environments (Ramas et al., 2023). Recent work shows that heutagogical approaches can strengthen learners' SDL and digital literacies, while also requiring shifts in teachers' roles towards mentoring and facilitating more flexible, learner-driven pathways (Ramas et al., 2023; Pretorius, 2025). In technology-mediated contexts, heutagogy has been used to design learning environments where students harness diverse digital tools to regulate their learning, collaborate and reflect, thereby creating a "heutagogical system" grounded in SDL (Xuan and Zhu, 2024). Importantly, we do not interpret heutagogy as teacher absence; rather, it implies a shift from transmission to facilitation and mentoring, in which teachers design conditions for agency while learners increasingly exercise choice and responsibility within curriculum constraints. In this study, we view TSCL as a practical means of enhancing SDL by introducing heutagogically informed experiences into Grade 8 mathematics classrooms. This was done by creating structured opportunities for learner agency, collaboration and reflection in an under-resourced, 4IR-driven context.

This study examined how a TSCL intervention in Grade 8 mathematics influences learners' perceptions of four SDL dimensions: learning motivation, planning and implementation, selfmonitoring and interpersonal communication (Shen et al., 2014). A mixed-methods design grounded within a DBR approach was used, integrating pre- and post-intervention inferential statistics, including effect size calculations using Cohen's *d*, with qualitative data from classroom observations and semi-structured interviews. Therefore, the primary research question of this study was: how does TSCL effect Grade 8 mathematics learners' perceptions of their SDL? While our study refines SDL theory by testing how TSCL affects motivation, planning, self-monitoring and interpersonal communication, it also offers a practical scaffold for teacher practice in schools. Although conducted over only two DBR cycles in a single regional context, our findings provide a model for scalable TSCL interventions that can be implemented in diverse settings. Although TSCL strengthens learner agency, the intervention was teacher-facilitated. Participating teachers received a workshop, codesigned lesson plans and materials with the researcher and were supported through classroom visits focused on reflective implementation.

2 Literature review

2.1 Persistent underperformance in international mathematics benchmarks

South Africa has prioritised STEM education as a driver of economic and social development (Chisom et al., 2024). STEM is broadly defined as an interdisciplinary framework that bridges academic concepts with real-world applications across science, technology, engineering and mathematics (Bybee, 2010). Consistent with its National Development Plan, the government aims to increase the number of learners who are strong in mathematics and science. The target is for about 450,000 school-leavers to qualify for bachelor's studies in STEM fields by 2030 (National Planning Commission, 2012). Yet despite these ambitions, mathematics performance remains alarmingly low. Widely debated internationally, South Africa is rated, for instance on TIMSS 2023 as performing among the lowest in terms of mathematics globally when measured against other countries (Kanjee et al., 2024). In fact, South Africa actually achieved the lowest primary grade mathematics achievement out of 64 participating countries and second lowest in the high school cycle among 44 countries (Von Davier et al., 2024). Table 1 shows Grade 9 learners' results in several countries, drawing on data from the 2007 to 2023 assessment cycles of TIMSS.

TABLE 1 Grade 9 learners' results in several countries, drawing on data from the 2007 until 2023 assessment cycles.

Year	2007	2011	2015	2019	2023
The country ranked first	Taiwan	South Korea	Singapore	Singapore	Singapore
Ranking of South Africa	Did not participate	Second last	Second last	Second last	Last
Number of participated countries	48	45	39	39	44

From the above table, it is clear that since 2007, South Africa has consistently ranked among the lowest performing countries in international mathematics assessments, and in the 2023 TIMSS cycle Grade 5 learners once again placed last out of 59 participating countries (Von Davier et al., 2024).

Although the 2023 data show a modest improvement for Grade 9 mathematics (from a scale score of 389 in 2019 to 397 in 2023), South Africa's relative global standing remains very weak. This underlines persistent systemic challenges faced by South African mathematics education. Equally concerning is that less than one in three matriculants study mathematics as a subject and only 50% of those who take it manage to pass (Kanjee et al., 2024). These results show a "maths crisis" that weakens the pathway into STEM fields (Prescott et al., 2020). Only a small number of learners qualify for advanced science and engineering programmes, and industries are reporting an increasing shortage of people with strong mathematical skills (National Youth Policy 2020, 2015:12).

These persistent challenges have deep roots in South Africa's schooling system. Lack of resources, perceptions teachers hold (their beliefs, opinions and views) about the subject and the legacy of apartheid-era separation are among other continuing factors, but so is classroom pedagogy (Tibane et al., 2024). In practice, many mathematics lessons are still teacher-centred and assessment-driven. Evidence suggests that teachers are under pressure to "teach to the test", leading them to concentrate on procedural rather than conceptual lessons (Venkat and Spaull, 2015:155–160). In addition to these challenges, the language barrier in a mathematics classroom is also a major problem for learners who are non-native English speakers. Pretorius (2015) argues that non-native English speakers struggle to understand mathematics concepts, the language used to phrase questions as well as to carry out procedures flexibly and accurately. Although English is used to teach and assess these learners, they may have little exposure to the language outside of the classroom (Pretorius, 2015).

One possible approach that can help resolve and thereby mitigate the issues of poor learner performance in mathematics is to enhance learners' perception of SDL ability (Bishara, 2021:93; Bailey and Lubbe, 2020:344). The idea that learners should become more self-directed towards their own learning would allow learners to think creatively and critically, keep up to date with content and become more committed to their learning (Morris, 2019:55). In this respect, this study aims to enhance mathematics learners' growth (in terms of SDL development) through the implementation of the TSCL strategy.

2.2 The imperative for self-directed learning in the fourth industrial revolution

The rapid acceleration of the Fourth Industrial Revolution (4IR) has intensified the call for learners to become self-directed (Briede and Popova, 2020). According to Knowles (1975, p. 18), SDL is "a process in which individuals take the initiative, with or without the help of others, in diagnosing their learning needs, formulating learning goals, identifying human and material resources for learning, choosing and implementing appropriate learning strategies, and evaluating learning outcomes." The integration of 4IR technologies has been recognised as a key enabler in fostering SDL among learners (Olivier, 2022). However, barriers such as limited access to digital resources and inadequate teacher training continue to hinder the effective integration of 4IR technologies in classrooms (Sehlako et al., 2023). These challenges underscore the need for targeted interventions that empower both teachers and learners with the necessary tools, competencies and support to participate meaningfully in digitally enriched learning environments.

Despite the centrality of SDL in 21st-century education, mathematics classrooms, particularly in lower-secondary grades, are dominated by a strong dependence on teacher-directed instruction that provides minimal opportunities for autonomy, peer discourse or reflective learning (Bishara, 2021). This traditional strategy frequently does not encourage learners in a profound way to become critical thinkers and problem solvers, which are must-have qualities needed for surviving in the 4IR era. At this level, learners begin engaging with abstract reasoning, non-routine problem solving and reflective monitoring (Schoenfeld, 2016), making SDL an essential component for successful mathematical learning. Recently, studies have emerged emphasising the benefits of blended learning environments, noting that the combination of digital tools and CL strategies strengthen learners' SDL abilities (Bosch and Pool, 2019; Laubscher and Bosch, 2025). Furthermore, Naidoo and Reddy (2023) established that technology-based teaching strategies, when effectively implemented, can enhance learner engagement and foster a deeper understanding of mathematical concepts.

Traditional approaches that focus predominantly on teaching formulae and routine procedures in mathematics have been criticised for their limited impact on developing learners' critical thinking and reasoning skills (Basri, 2019; Gitaari et al., 2013; Smith, 2014). Instead, mathematics education should provide learners with opportunities to ask meaningful questions, recognise patterns, formulate conjectures and make generalisations. These are practices that are fundamental to higher-order thinking and deep mathematical understanding (Basri, 2019). Research further suggests that improving learners' problem-solving abilities is vital for enhancing mathematics achievement (DBE, 2011) and that metacognitive skills, such as the ability to reflect on and regulate one's own thinking, play a key role in this process (Flavell, 1979; Hacker et al., 2009). Metacognition not only supports mathematical performance but is also essential for fostering lifelong learning capabilities (Jagals and der Walt, 2018). Therefore, learners must be equipped with the skills to search for, assimilate and integrate new information with prior knowledge, while also developing a positive disposition towards learning (Shen et al., 2014). Cultivating SDL competencies can therefore empower learners to take ownership of their learning journeys, maintain up-to-date content knowledge and improve their performance in mathematics (Bailey and Lubbe, 2020; Bishara, 2021).

Within this context, strengthening learners' SDL competencies becomes particularly important for mathematics. SDL has been identified as a key means of responding to continuous educational and societal change (Guglielmino, 2013; Lindberg et al., 2017) and can provide valuable insight into how to improve mathematics learners' performance and success. Enhancing mathematics learners' SDL abilities can assist them in gaining confidence, staying focused, acting as learning change agents and taking responsibility for making choices about their learning goals (Knowles, 1975). Self-directed learning development supports skills such as time management, note-taking, assignment and test planning and accepting the freedom to learn what is essential to them (Mentz and Van Zyl, 2016). Learners who see

themselves as makers of their own knowledge often experience greater motivation, as they gain a sense of autonomy in their studies (Van Zyl and Mentz, 2019). In this sense, the cumulative, conceptually demanding and problem-solving nature of mathematics makes selfregulated and autonomous engagement essential for sustained success in the subject.

While SDL is central to preparing learners for 21st-century demands, recent scholarship argues for extending this lens through heutagogy, or self-determined learning. Heutagogy emphasises learners' capacity to determine both the content and processes of their learning, encouraging flexible pathways and the development of capabilities rather than only competencies (Ramas et al., 2023). In a systematic review of empirical studies, Ramas et al. (2023) found that heutagogical approaches are associated with increased self-determined learning, stronger mentoring relationships and the cultivation of ICT skills. They also identified substantial challenges related to teachers' readiness, institutional cultures and wider environmental factors.

Heutagogy has primarily been explored in higher education and adult learning, yet its principles resonate strongly with the demands of lower-secondary mathematics, where learners must navigate abstract concepts, non-routine problems and sustained reflection. Pretorius (2025) illustrates how heutagogical principles, when combined with self-determination theory and active learning strategies, can support teachers and students to enhance agency, capability and metacognition in technology-enhanced environments. Xuan and Zhu (2024) show how English language learners use digital tools for self-paced exploration, reflection and collaboration within a heutagogically designed framework. These studies suggest that CL and TSCL can serve as powerful vehicles for enacting heutagogical principles, particularly when tasks explicitly require learners to plan, monitor and evaluate their own and peers' contributions. In the present study, we draw on this literature to conceptualise TSCL not only as a route to SDL, but also as an entry point into more self-determined, heutagogical forms of mathematics learning.

2.3 Cooperative learning: a strategy for enhancing learner engagement

Cooperative Learning offers a promising way to restructure classroom dynamics by fostering peermediated knowledge construction and mutual accountability (Johnson and Johnson, 2019). Across diverse contexts, including secondary school mathematics classrooms in Kenya (Maluni, 2021) and Pakistan (Javed et al., 2013), CL has consistently demonstrated significantly greater impacts on mathematics achievement compared to traditional teaching strategies. Evidence from South Africa further supports these findings. In a quasi-experimental study involving low-performing Grade 10 mathematics learners, Dhlamini and Mogari (2013) found that a group-based instructional strategy, where learners worked collaboratively to discuss and solve mathematical tasks, resulted in significantly improved performance relative to non-group-based approaches. This underscores the relevance and efficacy of collaborative learning within the national context.

Effective implementation of CL depends on five key principles identified by Johnson and Johnson (2009, 2013, 2019): positive interdependence, individual and group accountability, promotive interaction, social skills and group processing. These principles are essential to ensure meaningful collaboration and sustained learning gains. Positive interdependence encourages learners to rely on one another to achieve shared goals, while accountability ensures that each learner actively contributes and takes responsibility for their own and others' learning. Promotive interaction fosters mutual support, explanation and encouragement within the group. Social skills, such as communication, respect and cooperation, are necessary for productive group functioning and group processing provides opportunities for reflection, self-assessment and continuous improvement. When purposefully integrated and reinforced over time, these principles enhance not only academic performance but also important SDL-related capacities such as metacognition, motivation and collaborative goal setting (Mentz et al., 2008; Johnson and Johnson, 2016; Van der Merwe and Kruger, 2012).

Despite its benefits, research shows that CL in digital or blended settings can result in unequal participation, free-riding, and technology-related challenges, especially when learners have different levels of digital skills. This can negatively affect collaboration and learning outcomes (Dillenbourg, 1999; Järvelä et al., 2016; Johnson and Johnson, 2014; Kirschner et al., 2004). However, while the challenges and academic benefits of CL are well documented, its influence on SDL dimensions such as motivation, metacognitive regulation and collaborative goal-setting remains underexplored and warrants further investigation. In light of these limitations, the integration of digital tools into CL strategies offers a promising direction for strengthening learner autonomy and interaction, leading to the emergence of TSCL.

In this study, the link between CL and SDL is grounded in the idea that SDL is not solitary learning, but a process in which learners take initiative "with or without the help of others" (Knowles, 1975, p. 18). Cooperative learning provides structured conditions (positive interdependence, accountability, group processing and promotive interaction) that intentionally require learners to set goals, plan contributions, seek resources, evaluate progress and justify strategies to peers. These behaviours align directly with the four SDL dimensions measured in this study (motivation, planning and implementation, self-monitoring and interpersonal communication). Rather than positioning peers as regulators in place of learners, TSCL distributes support so that learners increasingly assume responsibility for decision-making and monitoring, while still benefiting from peer explanation, feedback and shared sense-making within structured tasks.

2.4 Technology-supported cooperative learning

The intersection between TSCL, that is, CL enhanced with digital tools (Johnson and Johnson, 1999), holds significant promise by combining the advantages of structured peer interaction with the dynamic capabilities of modern technology (Sekano et al., 2025). Recent research in CSCL for lower-secondary mathematics shows that using digital tools to connect learners and teachers can help them solve problems together and communicate better in online mathematics tasks, showing that CSCL principles are useful for designing TSCL in real classrooms (Van Hoe et al., 2024). TSCL facilitates structured peer collaboration, while integrating tools that enable reflection, planning and feedback. In the context of teacher

professional development, Sekano et al. (2023) demonstrated that TSCL can foster teachers' SDL when embedded within hands-on, practice-based training environments. Similarly, Mentz and van Zyl (2018) demonstrated the potential of CL to foster SDL in student teachers but did not examine this effect among school-aged mathematics learners. Various studies focus primarily on teacher professional growth within professional development or teachertraining contexts (Sebotsa et al., 2018; Verster et al., 2024), and not on how TSCL enhances learners' SDL in mathematics. One significant gap in the literature in this area is the shortage of theoretically grounded, systematic studies that are contex-specific and subjectfocused on adolescents' learning of mathematics.

While TSCL shares several characteristics with CSCL, this study deliberately adopts a cooperative learning framework as conceptualised by Johnson and Johnson (2009). We draw on their view of CL as structured small-group work designed around five core principles as mentioned previously: positive interdependence, individual and group accountability, promotive interaction, social skills and group processing. These principles guided the design of all group tasks in this intervention. As Lehtinen et al. (1999) argue, cooperation and collaboration differ in how participation is organised. In cooperative work, the task is divided so that individual learners are responsible for different parts of the problem, whereas collaboration involves joint, tightly co-ordinated engagement in solving the problem together. In contrast to traditional CL, CSCL emphasises continuous shared regulation and collaborative construction of knowledge (Ouyang and Zhang, 2024). By aligning TSCL with this perspective, our intervention harnesses both CL strategies and digital affordances to foster motivation, planning, self-monitoring and peer communication. In Grade 8 mathematics classrooms, we considered the structured division of tasks, clear role allocation and interdependence of CL to be more pedagogically appropriate than the more open-ended collaborative arrangements typically examined in CSCL research. In this study, TSCL implies that Grade 8 mathematics learners learn together in real classrooms with the help of digital tools, while CSCL is a wider field of research about how computers help people work and learn together in any setting (Stahl and Hakkarainen, 2021). While the theoretical benefits of TSCL are compelling, empirical research presents a more nuanced picture, highlighting both potential and limitations in different contexts.

2.5 Empirical studies on self-directed learning and cooperative learning with technology

Recent comparative research has shed light on how technology affects learners' perceptions of SDL and CL. For instance, Sui et al. (2024) used sophisticated modelling analyses to demonstrate how learners' self-efficacy and perceptions of the usability of self-assessment tools were significantly predictive of self-regulation in science learning. Lee et al. (2014) conducted a mixedmethod study into middle school learners' perceptions of personalised learning in both traditional and technology-enhanced learning environments. Interestingly, learner engagement in CL decreased with the introduction of technology and gender differences in SDL, which were usually in favour of adolescent girls. The value of having explicit self-assessment and self-monitoring tools was mentioned as a key means to increase self-regulation, suggesting that TSCL interventions in mathematics should incorporate such mechanisms to nurture motivation, planning and interpersonal communication, core elements of SDL. Despite increasing interest in TSCL, much of the existing literature focuses on teacher training or generalised implementations, with limited attention paid to how TSCL supports SDL development among school-aged learners in subject-specific contexts like mathematics. There is a need for contextually grounded, theoretically informed research that explores how adolescent learners experience and develop SDL competencies through TSCL in mathematics classrooms. This study addresses this gap by examining a TSCL intervention that integrates CL 305 strategies with digital tools, focusing specifically on Grade 8 learners and measuring changes across four SDL dimensions using a mixed-methods DBR design.

3 Methods

3.1 Research design

Viewed from a pragmatic paradigm, the study adopted a design-based research (DBR) approach, which is an appropriate method for the iterative development and empirical validation of an educational intervention in authentic educational settings (Design-Based Research Collective, 2003; McKenney and Reeves, 2025). A DBR approach was adopted because its iterative cycles allow for the collaborative co-design and refinement of the TSCL intervention within authentic classroom settings. This ensured that adjustments were driven by observed learner responses and teacher feedback. The iterative structure strengthens external validity by enabling the intervention to evolve in alignment with real-world constraints and pedagogical practices. Figure 1 shows a graphical representation of how the DBR process was realised in this study. The DBR process produced a more advanced model of TSCL over two intervention cycles, which could evaluate its effect on Grade 8 learners' perceptions of SDL in mathematics. The study followed a sequential explanatory mixed-method design and combined quantitative and qualitative data to generate robust and context-specific evidence.

The intervention was designed around the four fundamental CL principles: positive interdependence, individual accountability, face-to-face promotive interaction and group processing (Johnson and Johnson, 2013). In the experimental schools, these principles were implemented using pre-loaded digital content administered through tablets, self-paced quizzes, collaborative problem-solving activities and interactive tools, such as Mentimeter and GeoGebra. Learners also collaborated in structured group activities where each individual was assigned a specific role, simultaneously supporting autonomy and accountability. Tasks were aligned with the South African Grade 8 mathematics curriculum, which addressed parts of the curriculum on algebraic thinking, functions and number patterns. In contrast, the schools in the control group followed a traditional curriculum with the use of print workbooks, direct instruction and less collaborative interaction.

The intervention was implemented in two iterative cycles, each approximately six weeks long, to ensure consistency and pedagogical coherence. Before the implementation, teachers who participated from the experimental group completed a workshop on the pedagogy

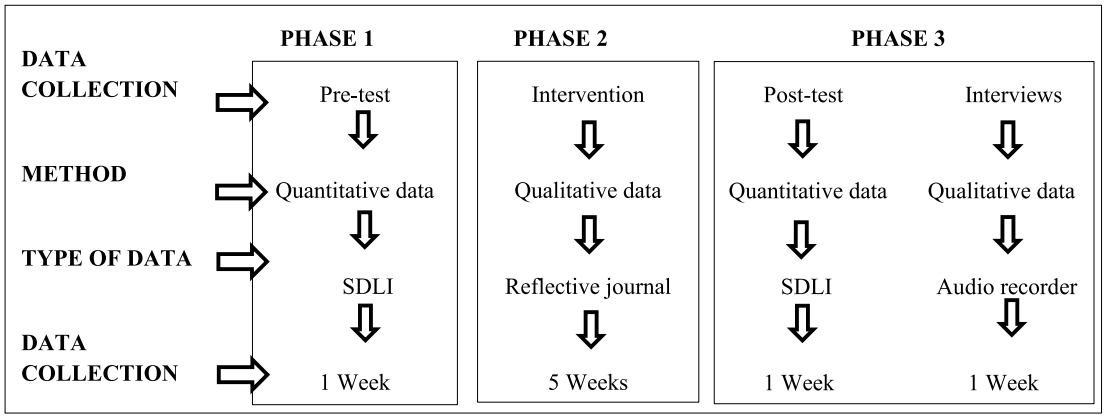


FIGURE 1
Graphical representation of the DBR process. Source: Author's own compilation.

of CL, integration of digital tools and classroom facilitation. Teacher lesson plans, guides and reflection templates were codesigned with teachers to situate the TSCL approach within their classroom contexts. Throughout the intervention, the researcher made some visits to the classrooms with the aim of offering nonevaluative support and encouraging reflective teaching. These scaffolds were created not only to improve learner SDL but also to increase teachers' confidence in facilitating TSCL in resourceconstrained settings. An illustration of the design of the TSCL intervention is presented in Figure 2 below.

3.2 Context and participants

This study was carried out in the Royal Bafokeng Nation (RBN) in South Africa, where there is an inequitable distribution of educational resources and differences in the availability of digital infrastructure. The RBN was selected because the researcher was already based within one of its secondary schools, allowing ongoing site access, continuous classroom engagement and feasibility for iterative DBR cycles. Ten public secondary schools offering Grade 8 mathematics were purposively sampled, with two schools comprising the experimental group (technology-enhanced) and eight schools serving as the control group (traditional textbook-based learning). Two schools were purposively assigned to the experimental group because they had already adopted a paperless classroom policy, including learner access to tablets, stable digital infrastructure and teachers with prior experience using digital tools for instruction and assessment. The remaining eight secondary schools, which did not have comparable digital infrastructure or routine technology use, were designated as the control group.

A total of $n = 427$ (i.e., $n = 101$ Experimental Group and $n = 326$ Control Group) Grade 8 learners participated in Cycle 1 and of $n = 522$ (i.e., $n = 131$ Experimental Group and $n = 391$ Control Group) learners in Cycle 2. Participant demographics included gender, prior exposure to digital tools and varying achievement levels in mathematics, ensuring that the sample reflected the diversity of learners and teachers within the RBN context. Learners in the experimental group used digital tablets, learning management systems and interactive platforms in mathematics learning. Teachers in participating schools also engaged in professional development to promote the use of TSCL strategies. Learners in the control group, by contrast, received conventional teacher-led mathematics instruction using printed workbooks and had no exposure to digital tools or structured CL strategies during the study period.

3.3 Data collection instruments

The study followed a sequential explanatory mixed-methods design in which quantitative SDLI pre- and post-measures informed the focus of qualitative classroom observations and interviews, allowing the qualitative strand to deepen and explain the quantitative trends. To quantitatively assess learners' perceptions of their SDL abilities, we used the SDLI questionnaire, which was developed by Shen et al. (2014). The SDLI has been tested across different populations and is designed according to four major SDL dimensions: motivation for learning, planning and implementation, self-monitoring and interpersonal communication. It is composed of Likert-scale items with a 5-point scale, ranging from "strongly disagree" to "strongly agree." It was administered to both the experimental and control groups before and after the intervention in each cycle. In the first iteration, 404 grade 8 mathematics learners out of the total sample of 427 completed the distributed pre-test SDLI questionnaires. A total of 404 questionnaires were distributed during the post-test phase of which 369 were completed. During the second iteration, 522 SDLI pre-test questionnaires were distributed and 513 were completed. A further 513 questionnaires were distributed during the post-test phase and 454 were completed. The SDLI was administered as a pre-test to establish baseline SDL levels for both the experimental and control groups. After completing the TSCL intervention, the same questionnaire was administered again to measure any changes in SDL abilities.

To triangulate the quantitative results and provide deeper contextual understanding, qualitative data was collected through semi-structured individual interviews and classroom observations. A purposive sample of 11 learners from the experimental group and four mathematics teachers participated in post-intervention interviews. These interviews explored their experiences, perceptions and reflections on the use of TSCL. In parallel, a total of 8 classroom observations (four per intervention cycle) were

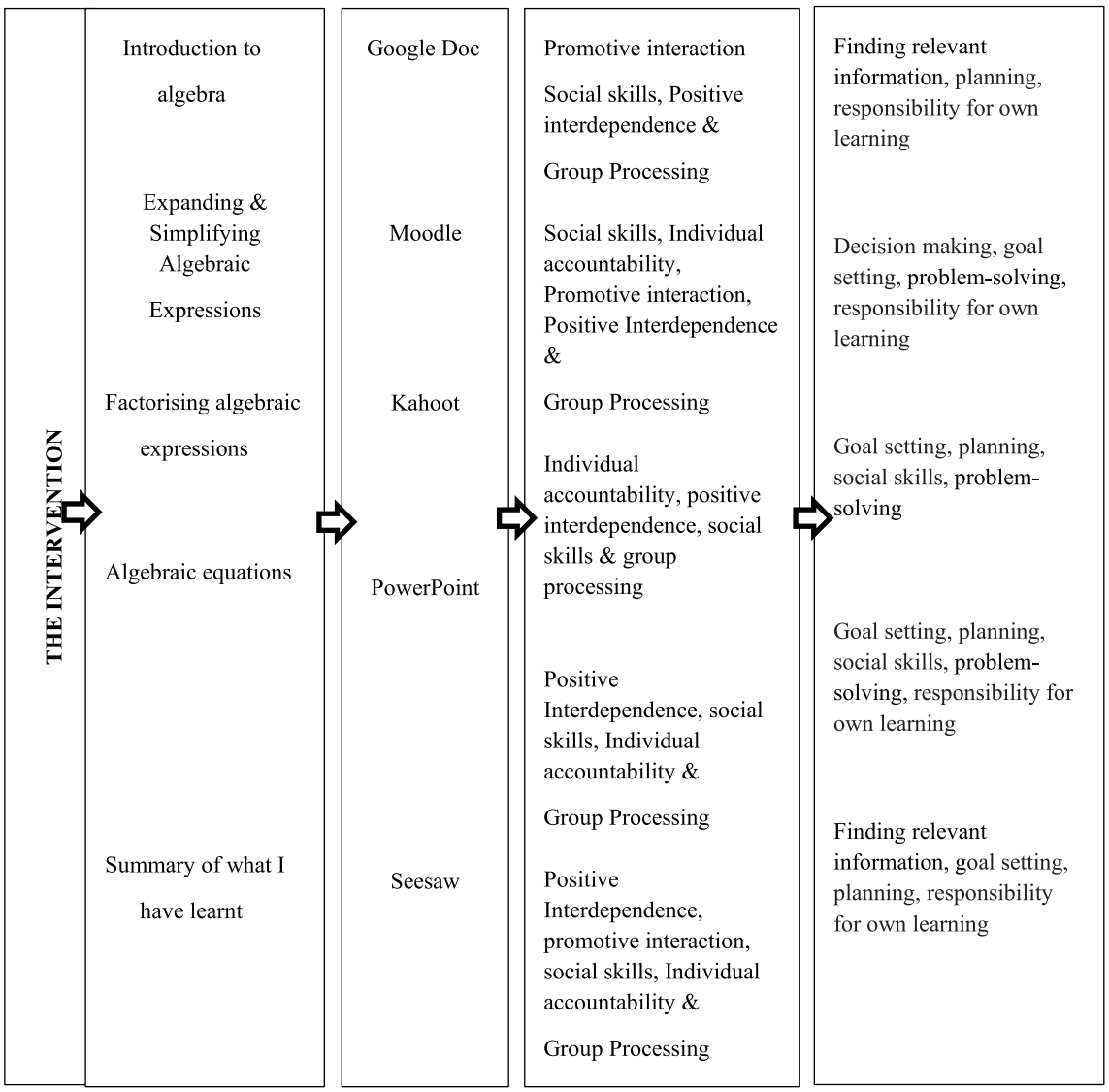


FIGURE 2
The technology-supported cooperative learning intervention design. Source: Author's own compilation.

conducted using a structured observation protocol. These observations provided rich, real-time data on learner interaction, the use of digital tools and the execution of collaborative tasks. The qualitative data collection aimed to capture learners' lived experiences with TSCL, particularly in relation to their sense of agency, motivation and engagement in mathematics learning. These findings were integral to interpreting the influence of TSCL on the development of SDL.

3.4 Data analysis

The initial stage of quantitative data analysis involved calculating descriptive statistics to examine baseline data distributions and compare pre- and post-intervention SDLI scores between the experimental and control groups. This provided an overview of learners' initial SDL profiles and enabled the identification of trends following the intervention. To assess whether changes were statistically significant, paired-sample t-tests were conducted within groups. These inferential tests compared pre- and post-intervention mean scores and were complemented by effect size calculations (Cohen's d) to evaluate the practical significance of the observed changes. Effect sizes were interpreted using Cohen's (1988) conventional thresholds: $d = 0.2$ (small), $d = 0.5$ (moderate) and $d = 0.8$ (large). This combined approach, involving statistical significance testing and effect size analysis, enhanced the robustness of the findings and reduced the risk of overlooking meaningful impacts of the intervention on learners'

perceptions of SDL. Results were reported across the four dimensions of SDL.

Qualitative data from interviews and classroom observations were analysed using inductive thematic coding, supported by ATLAS.ti. The analysis followed Braun and Clarke's (2006) six-phase approach to thematic analysis, beginning with familiarisation with the data, followed by the development of codes, initially generated both deductively, informed by SDL literature and inductively from the data itself. These codes were then organised into broader themes that captured learners' and teachers' experiences of participating in the TSCL intervention. To enhance analytical credibility, triangulation was employed: multiple researchers independently reviewed and interpreted the data before reaching consensus on the key themes. Peer debriefing sessions and iterative coding refinements were also conducted to ensure consistency and reduce potential researcher bias. This systematic process enabled a rich and credible interpretation of how TSCL shaped learners' motivation, autonomy and collaborative engagement in the mathematics classroom.

3.5 Ethical considerations

The study received ethical approval from the North-West University's Education Research Ethics Committee (Ref: NWU-00010-21-S2), ensuring that all procedures adhered to the institutional and national ethical standards for research involving human participants. Consent procedures were conducted in participants' preferred languages and assent was obtained from learners to ensure their voluntary participation and understanding of the research purpose. Participants were informed of their right to withdraw at any stage without penalty and data were securely stored for a five-year retention period. To uphold the privacy and the ethical standard and confidentiality were strictly maintained throughout the study. All interview transcripts and observational notes were anonymised and stored in password-protected virtual drives available only to the research team.

4 Findings

4.1 Quantitative results

Learners' perceptions of SDL were measured using the SDLI at the beginning and end of the intervention in both cycles. The SDLI includes four factors that assess learning motivation (LM), planning and execution (PI), self-monitoring (SM) and interpersonal communication (IC). Pairedsample *t*-tests and Cohen's *d* were used to investigate statistical and practical significance in the pre- and post-ATT score changes within the experimental group.

The results from cycle 1 (Table 2) revealed a statistically significant increase in performance in all the SDL areas. The largest effect sizes were found for Interpersonal Communication ($d = 0.66$) and.

Learning Motivation ($d = 0.63$), which could be categorised as moderate. The effects on Planning and Implementation ($d = 0.58$) and Self-Monitoring ($d = 0.45$) were also significant showing that the learners became more autonomous and reflective during their learning after the intervention.

In Cycle 2, with a larger cohort ($n = 205$), the pattern of improvement persisted across all SDL dimensions (Table 3). Again, Interpersonal Communication ($d = 0.69$) and Learning Motivation ($d = 0.61$) represented the most pronounced improvements in collaborative and intrinsic learner growth. There were also moderate effects for Planning and Implementation ($d = 0.60$) and Self-Monitoring ($d = 0.44$).

These trends are graphically shown in Figure 3, which illustrates pre- and post-test SDL scores for each SDL dimension in both cycles. The figure clearly shows increasing shifts in the perception of learners after the intervention of the TSCL.

The control group, which received traditional, teacher-directed instruction without the integration of technology or CL strategies, showed no statistically significant change in any of the SDL dimensions across either cycle. This lack of change suggests a discernible effect of the TSCL intervention on the development of learners' SDL abilities in the experimental group.

TABLE 2 Pre- and post-test SDL scores for experimental group—Cycle 1.

SDL dimension	Pre-test mean (SD)	Post-test mean (SD)	Mean difference	*p*-value	Cohen's *d*
Learning motivation	3.12 (0.51)	3.67 (0.53)	+0.55	<0.001	0.63 *(moderate)*
Planning and implementation	3.05 (0.49)	3.59 (0.50)	+0.54	<0.001	0.58 *(moderate)*
Monitoring	2.97 (0.52)	3.34 (0.56)	+0.37	0.004	0.45 *(small–moderate)*
Interpersonal communication	3.18 (0.58)	3.80 (0.60)	+0.62	<0.001	0.66 *(moderate)*

TABLE 3 Pre- and post-test SDL scores for experimental group—Cycle 2.

SDL Dimension	Pre-test mean (SD)	Post-test mean (SD)	Mean difference	*p*-value	Cohen's *d*
Learning motivation	3.20 (0.50)	3.76 (0.52)	+0.56	<0.001	0.61 *(moderate)*
Planning and implementation	3.09 (0.47)	3.66 (0.49)	+0.57	<0.001	0.60 *(moderate)*
Monitoring	3.02 (0.54)	3.41 (0.56)	+0.39	0.003	0.44 *(small– moderate)*
Interpersonal communication	3.22 (0.59)	3.91 (0.63)	+0.69	<0.001	0.69 *(moderate)*

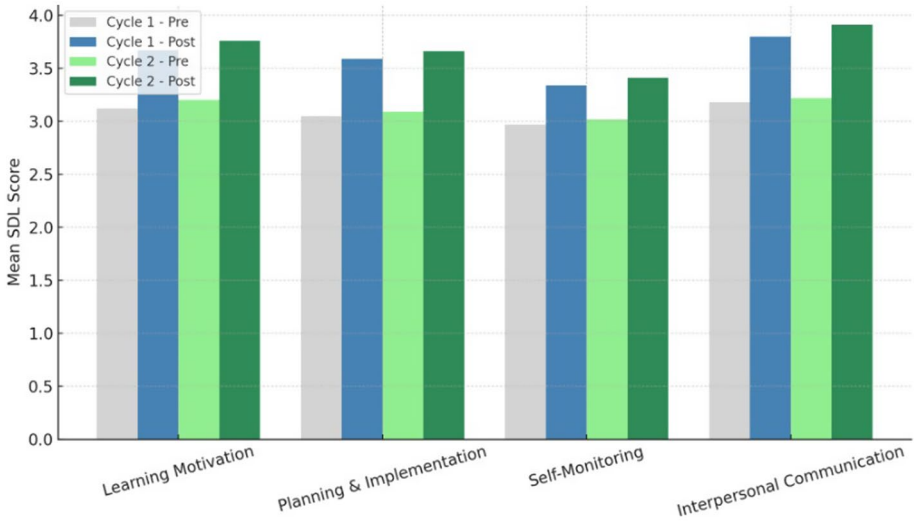


FIGURE 3
Pre- and post-test SDL scores by cycle and dimensions. Source: Author's own compilation.

4.2 Qualitative findings

To triangulate the quantitative findings and provide deeper insight into learners' and teachers' experiences with the TSCL intervention, qualitative data was collected through semi-structured interviews and classroom observations. Participants' identities remain confidential and quotes are accompanied by identification codes: (T#:#) for teachers and (L#:#) for learners, indicating transcript and line number. Four interrelated themes emerged from the data: (1) Technology as a catalyst for engagement and personalised learning; (2) CL strategies supporting SDL development; (3) Emergence of core SDL behaviours and skills; and (4) Teacher perspectives on implementation and required support.

4.2.1 Theme 1: Technology as a catalyst for engagement and personalised learning

Learners described the integration of digital tools, such as Seesaw, Moodle and GeoGebra, as engaging and motivating. Technology afforded opportunities for personal pacing, creative expression and selfassessment.

> *"Recording our voices, taking pictures, and writing on them using Seesaw, was different compared to what we normally do in our classes every day." (L15:3).*
>
> *"When I was at home … I was learning at my own speed and I could even skip ahead if I have to." (L7:3).*

Teachers observed that the technology-based intervention facilitated greater learner participation:

> *"… inquiry-based instruction encourages my learners to ask questions and investigate their own ideas …" (T1:7).*

However, learners also identified challenges, including data access, poor connectivity and difficulties during virtual collaboration:

> *"Where I stay, the connection is bad." (L3:2).*
>
> *"Some members will bluetick you." (L16:4).*

Despite these barriers, most learners expressed a desire for continued exposure to TSCL:

> "Sir … are we going to have other activities after this?" (L6:13).

4.2.2 Theme 2: Cooperative learning to support SDL development

TSCL provided a structured environment where peer interaction, accountability and shared responsibilities fostered SDL skills. Learners reported a growing reliance on one another for understanding and task completion:

> *"… because I knew that some members in our group are good with designing and presentation skills, I made sure that I all the information we need is readily available, otherwise I wouldn't be able to complete the task on my own." (L13:3).*

Learners demonstrated understanding of core CL principles, including positive interdependence, individual accountability, promotive interaction, group processing and social skills: "... … we had to count on our different abilities …" *(L16:13).*

> *"On the last day of submission, we would check and edit each other's work, to make sure that everything is in order." (L8:14).*
>
> *"We always motivated each other to work hard." (L16:9).*

> *"One of our main ground rules was that after a person finished speaking, someone has to rephrase what was said to make sure that everyone understands it." (L8:14).*

> *"We listened carefully to each member's ideas or comments before responding." (L9:15).*

4.2.3 Theme 3: The emergence of core SDL characteristics and skills

Learners demonstrated increased agency, goal setting, monitoring and evaluating their own learning process; having a strong ability to learn independently; finding joy in learning, organising and planning, as the intervention progressed:

> *"Me and my team were very organised. We made sure that we set an appointment for catch-up to make sure everything is ready before submission." (L6:10).*

> *"What helped us in our group to succeed is that we always divided tasks according to our strengths." (L11:17).*

Several participants shared how they took responsibility for their own learning and managed setbacks:

> *"When I was doing my section of work … I tried to visit many different websites." (L6:9).*

> *"I am proud of the amount of information I provided, even when I find the task difficult." (L4:14).*

4.2.4 Theme 4: Teacher perspectives on implementation and required support

Teachers highlighted the value of TSCL in promoting learner engagement but also expressed the need for professional development and ongoing support, particularly in integrating diverse technologies and implementing structured CL:

> *"I saw how they were excited about using different apps other than Moodle … I think it is something worth considering for my lessons too." (T1:12).*

Many teachers admitted to a limited understanding of CL strategies and expressed concern over classroom management and curriculum coverage:

> *"Since our curriculum changed to being online only, it is difficult for me to use cooperative learning." (T3:7).*

> *"I want to learn more about how to expose them to a different style of learning with different cognitive demands." (T3:10).*

The establishment of a community of practice (CoP) was proposed by teachers as a sustainable support strategy:

> *"We know exactly what we are struggling with … I think we can meet regularly, perhaps every Wednesday, as Maths subject teachers." (T5:11).*

Table 4 provides an integrated summary linking each SDL dimension with representative qualitative excerpts and interpretive insights, demonstrating how learner experiences aligned with the quantitative shifts observed.

The qualitative findings enrich the quantitative results by showing how the observed gains in SDL dimensions were enacted in learners' and teachers' day-to-day experiences. Statistically significant improvements in learning motivation, planning and implementation, self-monitoring and interpersonal communication corresponded with learners' descriptions of increased interest in mathematics, clearer division of roles in groups, greater use of digital feedback to check their work and more frequent peer explanation and support. At the same time, the more modest gains in self-monitoring align with accounts of learners still relying on technological and peer prompts and with teachers' recognition that metacognitive regulation requires further scaffolding. As summarised in Table 4, the quantitative and qualitative strands thus offer a coherent picture of how TSCL, underpinned by explicit cooperative structures and purposeful use of technology, supports the development of SDL in Grade 8 mathematics. It also highlights areas where more sustained intervention and teacher professional learning remain necessary.

TABLE 4 Integrated summary of SDL dimensions with representative qualitative evidence and interpretive insights.

SDL dimension	Quantitative change (pre-post/effect size)	Representative qualitative evidence (quotes, observations)	Interpretation
Motivation	Mean score (prepost), Cohen's $d = 0.52$	"I felt more interested in maths now that we work together and use the app." (*L7:11*)	TSCL fostered increased learner engagement and internal motivation through social accountability and interactive tasks.
Planning and implementation	Significant prepost gain, $d = 0.44$	"We first discussed how we will approach the problem, then divided tasks among us." (*L16:7*)	Learners developed better planning strategies and task coordination under cooperative, digitally mediated conditions.
Self-monitoring	Pre-post increase, $d = 0.61$	"The feedback from the tool helped me see where I went wrong and correct it." (*L3:10*)	Digital feedback and peer review supported metacognitive regulation and improved self-assessment behaviours.
Interpersonal communication	Moderate gain, $d = 0.48$	"We explained solutions to each other and debated why one method works better." (*L10:14*)	TSCL enhanced peer dialogue, argumentation and collective reasoning—key for collaborative SDL development.

5 Discussion

This study investigated the extent to which TSCL influenced Grade 8 learners' perceptions of their SDL abilities in mathematics. Drawing on both quantitative and qualitative data collected across two iterative intervention cycles, the findings demonstrate how TSCL contributed meaningfully to the development of SDL. This development was explored particularly in the domains of learning motivation, planning and implementation and interpersonal communication. The discussion also situates the findings within heutagogical perspectives and 4IR learning demands, demonstrating how TSCL supports learner agency in technology-supported environments.

Quantitative results across both intervention cycles indicated statistically and practically significant improvements in learners' perceptions of their SDL abilities, particularly in the domains of learning motivation, planning and implementation and interpersonal communication. The moderate effect sizes observed for these dimensions suggest that TSCL offers more than marginal improvements. It supports substantive shifts in how learners engage with mathematical content and with one another. These numerical shifts were mirrored in learners' descriptions of their experiences. For example, learners linked their increased interest in mathematics directly to the use of apps and group work ("I felt more interested in maths now that we work together and use the app"), while others highlighted how they planned their approach and divided tasks according to group members' strengths. Classroom observations further showed frequent peer explanation and debate around solution strategies, signalling stronger interpersonal communication and collaborative problem-solving. These findings align with earlier research (Shen et al., 2014; Mentz and Bailey, 2019), affirming that structured CL can promote engagement, agency and collaborative learning behaviours. As observed from the SDLI data, the improvements in the SDL dimensions can be read as early movement towards more self-determined forms of learning. As Ramas et al. (2023) argue, heutagogy aims to cultivate learners who can set their own goals, regulate their learning processes and transfer knowledge flexibly to new contexts. In our TSCL intervention, learners increasingly took responsibility for group roles, monitored their own and peers' progress and used digital tools to revisit content and track understanding. These are behaviours that align with the capability-building and double-loop learning emphasised in heutagogical frameworks. The combination of moderate gains in motivation and planning ($d \approx 0.6$ in both cycles) with learners' accounts of setting appointments, working ahead at their own pace and seeking additional online resources suggests emerging self-determined behaviour rather than mere compliance. In this sense, the TSCL tasks operated as "heutagogical moments" where learners exercised choice, negotiated roles and reflected on progress within the constraints of the Grade 8 curriculum.

Although improvements in self-monitoring were observed, effect sizes were slightly smaller compared to the other SDL dimensions. This finding is consistent with previous studies (Loyens et al., 2008; Thornton, 2010), which suggest that metacognitive abilities such as monitoring and regulating one's own learning processes may require longer-term intervention and more explicit scaffolding than is often provided in short cycles.

Qualitative findings added explanatory depth to the quantitative outcomes. Learners described how digital tools facilitated engagement, supported autonomy and enabled them to access learning materials beyond the classroom. The use of platforms such as Moodle, Seesaw and GeoGebra empowered learners to take initiative, reflect on their understanding and communicate more confidently with their peers. These insights affirm the role of technology as a cognitive and motivational scaffold for SDL. The nature of the CL activities in the intervention appeared to contribute to the development of SDL by fostering peer accountability, shared responsibility and cooperative problem-solving. Learners not only supported one another's understanding but also developed critical interpersonal and communication skills. These findings reinforce theoretical perspectives such as those given by Johnson and Johnson (2013) and Gillies (2007) that view CL as a powerful mechanism for nurturing SDL, especially when learners are assigned meaningful roles. As summarised in Table 4, each SDL dimension showing quantitative improvement is associated with concrete learner behaviours and teacher observations, illustrating how TSCL translated into lived classroom practices.

We position technology in TSCL as a scaffold for SDL rather than a substitute teacher. Digital tools contributed prompts, immediate feedback and accessible resources that learners could use to check understanding and revisit content, while teacher facilitation and CL principles shaped how these affordances were taken up. At the same time, technology is not neutral: learners reported connectivity barriers and uneven participation online. Prior studies caution that technology can dampen collaborative engagement in some contexts. For this reason, future iterations of TSCL should purposefully combine digital and non-digital resources, selecting technology where it strengthens feedback, pacing and representation, while retaining low-tech and concrete materials for tasks where they better support sense-making and sustained attention.

Teachers' reflections provided further insight into the implementation process. While many reported that TSCL encouraged greater learner engagement and autonomy, they also acknowledged limitations in their own understanding of CL strategy. This resonates with the quantitative pattern where the control group who taught through more traditional, teacher-centred approaches, showed no significant change in any SDL dimension. Teachers' uncertainty about how to move beyond conventional group work towards structured CL suggests that, without sustained professional development, it is unlikely that similar SDL gains would emerge in non-intervention classrooms.

Several teachers equated group work with CL, but did not fully implement the core principles of CL, necessary for effective TSCL. This highlights the need for ongoing professional development, including not only the use of digital tools but also the pedagogy of structured collaboration, classroom management in digital contexts and the facilitation of learner agency. The study's findings also speak to the role of teachers in designing and sustaining heutagogically oriented environments. Pretorius (2025) demonstrates that, when teachers adopt active learning approaches informed by selfdetermination theory and heutagogy, they shift from transmitting content to facilitating learners' selfdetermined engagement with disciplinary ideas. In our study, mathematics teachers who implemented TSCL were required to re-conceptualise their roles: they co-designed digital tasks, negotiated group structures and provided ongoing formative feedback rather than controlling every step of the learning process. The study suggests that TSCL can act as a bridge between traditional, teacher-centred mathematics teaching and more heutagogically oriented, self-determined learning environments for adolescents. The contributions of this study are articulated across theoretical, methodological and practical dimensions, showing how the work advances SDL theory, illustrates the applicability of DBR in Grade 8

mathematics and offers actionable guidance for teachers implementing TSCL in resource-constrained contexts.

6 Practical actionable implications for teaching and curriculum design

In line with the work of Sekano et al. (2023) where guidelines to implement TSCL in teacher professional development were discussed, this study revealed the following additional implications for teaching and curriculum design. The following set of implementation principles for TSCL in Grade 8 mathematics classrooms were evident from the findings: (1) incorporate the deliberate structuring of CL, by intentionally stimulating the five basic principles of CL, (2) as guided by heutagogical principles, purposefully select digital tools that support planning, monitoring and feedback to enhance SDL, (3) explicitly integrate SDL skills while scaffolding mathematical content and (4) ensure the alignment of cooperative tasks with curriculum goals and assessment practices.

7 Limitation and future research

This study's findings should be interpreted in light of several limitations. First, although the research was not conducted within a typically resource-constrained context, the TSCL intervention was implemented over a relatively short period in only two paperless secondary schools in one district, that were comparatively well positioned in terms of digital infrastructure and teacher readiness. This limits the generalisability of the findings, particularly to schools with minimal access to digital devices, unstable connectivity, or limited teacher experience with educational technologies. Second, participation was voluntary and the sample may therefore reflect a degree of self-selection bias. Future research should investigate TSCL through longer-term, multi-site studies in which more teachers implement the intervention over extended periods, to test the sustainability and transferability of SDL gains. There is also a need for ongoing, well-resourced professional development to support mathematics teachers in designing and facilitating TSCL as a routine part of their practice.

8 Conclusion

This study investigated the potential of TSCL to enhance Grade 8 learners' SDL abilities in mathematics. This study reaffirmed its purpose to examine how TSCL influences Grade 8 learners'.

SDL perceptions. Effect sizes ranged from $d = 0.44$ to 0.69, demonstrating moderate to strong intervention effects. The integration of digital tools with CL principles improved learners' perceptions of their SDL, particularly in motivation, planning and interpersonal communication. Qualitative findings corroborated these results, offering insight into how learners developed autonomy, collaboration and reflective thinking. The study also underscored the importance of teacher professional development to support effective TSCL implementation. The findings suggest that TSCL presents a viable pedagogical approach for fostering SDL in mathematics classrooms and addressing the demands of 21st-century learning.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author/s.

Ethics statement

The studies involving humans were approved by The North-West University Education, Management and Economic Sciences, Law, Theology, Engineering and Natural Sciences Research Ethics Office (NWU-EMELTEN-REC). The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation in this study was provided by the participants' legal guardians/next of kin.

Author contributions

GS: Formal analysis, Writing – original draft, Conceptualization, Writing – review & editing, Methodology, Validation. DL: Supervision, Writing – review & editing. RB: Supervision, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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